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Published: 10 November 2025

Citation in BibTeX format

HAI '25: International Conference on Human-Agent Interaction
November 10 - 13, 2025
Yokohama, Japan

POSTER

AI-Enhanced Interactive Storytelling: Supporting Executive Functions and Cultural Learning Through Participatory Design

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Open Access Support provided by:

Umeå University

HAI '25: Proceedings of the 13th International Conference on Human-Agent Interaction (November 2025)
https://doi.org/10.1145/3765766.3765806
ISBN: 9798400721786

AI-Enhanced Interactive Storytelling: Supporting Executive Functions and Cultural Learning Through Participatory Design

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Abstract

This paper explores the co-design of an Interactive Digital Narrative (IDN) aimed at supporting the development of executive function (EF) in young children through culturally grounded interactive storytelling. Adopting a user-centered design approach, the research followed a five-phase iterative process involving nine participants from various disciplines, including social sciences, education, and Artificial Intelligence (AI), as well as parents. Three co-design workshops were conducted: (1) to integrate Swedish culture and values into the IDN, (2) to embed EF-stimulating interactions based on parent-child play, and (3) to conceptualise an AI-enhanced support tool for parent-child storytelling and cognitive development. Thematic analysis was employed to extract design insights from the audio recording of the workshops. These insights informed the development of a high-fidelity prototype featuring culturally resonant story elements that can support the development of EF in young children. Additionally, by positioning parents as companions rather than instructors and by embedding EF-supportive design into a joyful storytelling context, this work provides a framework for designing IDNs that are both pedagogically effective and playful for young children. This paper presents initial results of co-design of this IDN.

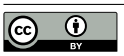
CCS Concepts

• **User Centered Design**; • **Participatory Design**; • **Human Computer Interaction (HCI)**;

Keywords

Interactive Digital Narrative, Executive Functions, Co-Design, Swedish Culture, Human-Computer Interaction, Artificial Intelligence

ACM Reference Format:
Gabriele Leonelli and Arzu Guneysu. 2025. AI-Enhanced Interactive Storytelling: Supporting Executive Functions and Cultural Learning Through Participatory Design. In *13th International Conference on Human-Agent Interaction (HAI '25), November 10–13, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3765766.3765806



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HAI '25, Yokohama, Japan
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ACM ISBN 979-8-4007-2178-6/2025/11
https://doi.org/10.1145/3765766.3765806

1 Introduction

Storytelling is central to how humans transmit culture, values, and traditions. In the digital era, Interactive Digital Narratives (IDNs) allow users to influence the progression of a story, creating participatory and immersive experiences [1, 18]. Beyond entertainment, IDNs offer opportunities for education, creativity, and socio-cultural learning [19].

This project investigates how an IDN can be co-designed to both represent Swedish culture and foster executive functions (EF) in young children. EF refers to higher-order cognitive processes such as inhibitory control, working memory, and cognitive flexibility [11]. These skills are predictive of school readiness, long-term health, and social well-being [2, 14]. Stimulating EF during early childhood, especially in engaging and socially meaningful contexts such as parent–child play, may therefore have enduring benefits.

The study further explores how Artificial Intelligence (AI) can enrich interactive storytelling. AI companions embedded in IDNs have the potential to scaffold reflection, introduce new challenges, and adapt narratives to user needs.

The central research questions are:

1. How can an IDN be co-designed to represent Swedish folklore while fostering cultural engagement?
2. How can IDN interactions support EF development in children through parent–child engagement?
3. How can AI features be designed to scaffold both storytelling and cognitive stimulation?

2 Background

IDNs combine narrative and interaction, allowing users to influence how a story unfolds. They have been used not only for entertainment but also for education and creativity, showing potential to foster social interaction and cross-cultural learning [18, 19]. Prior work, such as the Mobeybou project, demonstrated how cross-cultural IDNs can support early literacy while encouraging collaboration and imagination [21]. Importantly, these kinds of interactive experiences can also engage higher-order cognitive processes, including planning, rule-following, and problem-solving, thereby indirectly stimulating the development of executive functions [12].

EF are central to cognitive and socio-emotional development in early childhood, comprising skills such as working memory, inhibitory control, and cognitive flexibility [9]. These skills emerge rapidly between the ages of three and six and form the foundation for later learning and self-regulation [6, 8]. Research has shown that interventions embedded in playful, socially meaningful contexts, such as parent–child interaction, are especially effective in



Figure 1: Mood board to represent the Swedish culture.

promoting EF growth [10, 15]. Embedding EF training within culturally resonant storytelling thus represents an opportunity to make cognitive support both engaging and accessible.

At the same time, developments in Artificial Intelligence (AI) create new opportunities for adaptive and supportive storytelling companions. AI can scaffold reflection, pose challenges, or introduce new narrative directions that enrich user engagement [7, 17]. Within the context of IDNs, AI agents can be designed not simply as technical systems but as narrative partners that help children and parents co-create stories, encouraging both cognitive stimulation and cultural connection.

3 Methodology

This study employed a qualitative, constructivist, and interpretive approach grounded in user-centered design and co-creation principles [13, 22] User-centered and co-creation principles guided the research [16, 20], supporting the development of a culturally resonant and cognitively enriching digital experience for children.

3.1 Data Collection and Analysis

Data were gathered through audio-recorded workshops, participant-generated artefacts (eg, sketches, sticky notes), and a mixed-methods usability questionnaire. Reflexive thematic analysis [3–5] informed successive iterations of the prototype.

3.2 Five-Phase Design Process

The process involved nine participants, students, researchers, and parents, across five phases:

1. Ideation – Literature review, brainstorming, and creation of initial mock-ups to identify characters, landscapes, and objects (see fig. 1 and 2).
2. Cultural Design Workshop – Four social science students co-designed narrative elements inspired by Swedish folklore.
3. EF Workshop – Two participants, including parents, proposed activities to stimulate EF during parent–child storytelling.
4. AI and Parents Workshop – Three participants with AI expertise and parents designed features to scaffold narrative engagement, like the Wise-AI and the Aurora.
5. Prototype Evaluation – Seven participants tested a high-fidelity prototype in Figma, completing usability tasks and questionnaires.

4 Results

4.1 Cultural Representation

The first workshop revealed three key themes:

- Pop literature and mythology: Astrid Lindgren's works, trolls, and runestones were viewed as essential cultural anchors.
- Traditional festival: Midsommar celebrations and summer landscapes were seen as quintessentially Swedish.
- Nature and animals: Forests were described as spiritual spaces; animals such as bears were considered morally complex symbols.

These insights informed the design of story settings, protagonists, and antagonists.

4.2 Executive Function Interactions

Participants proposed parent–child activities, such as planning story choices together, role-playing characters with rules and constraints, and responding to narrative twists introduced mid-story. These concepts were developed to stimulate EF. These ideas shaped scenarios and interaction flows in the prototype.

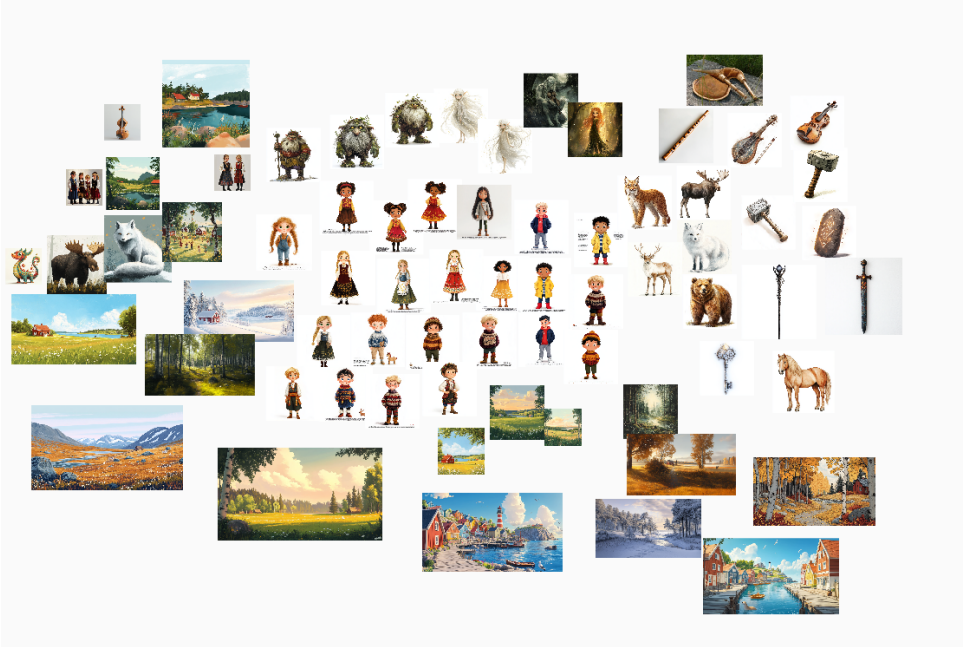


Figure 2: Mock-ups of the possible elements that will represent the Swedish culture.

4.3 AI Features

Two AI features emerged. Wise-AI: Acts as a reflective guide, asking questions like *"What might happen if the troll takes the stone?"* to prompt planning and perspective-taking. Aurora: Introduces culturally grounded narrative twists, stimulating creativity and cognitive flexibility.

Both were represented through Figma interaction flows, though not technically implemented.

4.4 Prototype Evaluation

Seven participants tested the high-fidelity prototype.

- Usability and engagement: Most found the system enjoyable and culturally authentic. One participant remarked, *"It feels like a real Swedish story, but more playful."*
- Perceived value: Wise-AI was appreciated for guiding reflection, but its role needed clearer definition. Aurora was inspiring but occasionally disruptive, highlighting the need for adjustable intervention.
- EF support: Participants recognized the potential for stimulating memory and planning but emphasized the importance of future testing with children.

5 Conclusion

This study demonstrates how participatory design can integrate cultural narratives, executive function development, and AI-supported interaction into an IDN. The co-creation of story elements inspired by Swedish folklore provided authenticity and cultural resonance, indicating that grounding interactive storytelling in familiar traditions and myths can deepen engagement and foster a sense of belonging for families. At the same time, embedding activities that target executive functions illustrated the potential of playful and collaborative storytelling to support planning, inhibition, and cognitive flexibility. Participants described how decision-making, role-playing, and responding to narrative challenges could create meaningful opportunities for children to practice these skills in ways that felt both natural and enjoyable.

The design of the AI features offered further insights into how technology might enrich these experiences. Wise-AI was envisioned as a reflective companion that encouraged pausing and considering alternative perspectives, while the Aurora feature introduced culturally grounded narrative twists that added novelty and imaginative depth. The evaluation highlighted the promise of these roles, but it also revealed the need for clearer scaffolding to balance supportive guidance with the risk of disrupting the flow of storytelling. This highlights the importance of carefully calibrating when and how AI interventions occur.

Although the findings are encouraging, limitations must be acknowledged. The evaluation involved a small group of adult participants, rather than children and parents, in everyday contexts, which limits generalizability. Furthermore, the AI components were developed as conceptual prototypes rather than fully implemented systems. These constraints limit strong claims about the effectiveness of supporting executive functions or sustaining long-term engagement.

Despite these limitations, the project makes a significant contribution to research on human–agent interaction and interactive storytelling by presenting a design pathway that integrates cultural traditions, cognitive development, and AI. It demonstrates that IDNs can serve as a platform for embedding executive function training in culturally meaningful contexts, and that AI can be conceptualized

not only as a technical system but also as a narrative partner. Future work will involve testing the prototype with families, refining the scaffolding roles of Wise-AI and Aurora, and assessing long-term cognitive and cultural outcomes. Taken together, the study provides initial but valuable evidence that culturally grounded, AI-enhanced IDNs hold promise as engaging and educational tools that support both storytelling and the development of children's executive functions.

References

[1] Mattia Bellini. 2022. Interactive Digital Narratives as Complex Expressive Means. *Front. Virtual Real.* 3, (May 2022), 854960. https://doi.org/10.3389/frvir.2022.854960

[2] Clancy Blair and Rachel Peters Razza. 2007. Relating Effortful Control, Executive Function, and False Belief Understanding to Emerging Math and Literacy Ability in Kindergarten. *Child Development* 78, 2 (March 2007), 647–663. https://doi.org/10.1111/j.1467-8624.2007.01019.x

[3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (January 2006), 77–101. https://doi.org/10.1191/1478088706qp063oa

[4] Virginia Braun and Victoria Clarke. 2021. Can I use TA? Should I use TA? Should I *not* use TA? Comparing reflexive thematic analysis and other pattern-based qualitative analytic approaches. *Couns and Psychother Res* 21, 1 (March 2021), 37–47. https://doi.org/10.1002/capr.12360

[5] David Byrne. 2022. A worked example of Braun and Clarke's approach to reflexive thematic analysis. *Qual Quant* 56, 3 (June 2022), 1391–1412. https://doi.org/10.1007/s11135-021-01182-y

[6] Kelly B. Cartwright. 2012. Insights From Cognitive Neuroscience: The Importance of Executive Function for Early Reading Development and Education. *Early Education and Development* 23, 1 (January 2012), 24–36. https://doi.org/10.1080/10409289.2011.615025

[7] Sarah A. Chauncey and H. Patricia McKenna. 2023. An Exploration of the Potential of Large Language Models to Enable Cognitive Flexibility in AI-Augmented Learning Environments. In *Proceedings of the Future Technologies Conference (FTC) 2023, Volume 4*, Kohei Arai (ed.). Springer Nature Switzerland, Cham, 135–153. https://doi.org/10.1007/978-3-031-47448-4_11

[8] Matthew C. Davidson, Dima Amso, Loren Cruess Anderson, and Adele Diamond. 2006. Development of cognitive control and executive functions from 4 to 13 years: Evidence from manipulations of memory, inhibition, and task switching. *Neuropsychologia* 44, 11 (January 2006), 2037–2078. https://doi.org/10.1016/j.neuropsychologia.2006.02.006

[9] ADELE Diamond. 2016. Why improving and assessing executive functions early in life is critical. (2016).

[10] Adele Diamond and Kathleen Lee. 2011. Interventions Shown to Aid Executive Function Development in Children 4 to 12 Years Old. *Science* 333, 6045 (August 2011), 959–964. https://doi.org/10.1126/science.1204529

[11] Adele Diamond and Daphne S. Ling. 2019. Review of the Evidence on, and Fundamental Questions About, Efforts to Improve Executive Functions, Including Working Memory. In *Cognitive and Working Memory Training* (1st ed.), Jared M. Novick, Michael F. Bunting, Michael R. Dougherty and Randall W. Engle (eds.). Oxford University PressNew York, 143–431. https://doi.org/10.1093/oso/9780199974467.003.0008

[12] Maitê Gil and Cristina Sylla. 2022. A close look into the storytelling process: The procedural nature of interactive digital narratives as learning opportunity. *Entertainment Computing* 41, (March 2022), 100466. https://doi.org/10.1016/j.entcom.2021.100466

[13] Cheu-Jey George Lee. 2012. Reconsidering Constructivism in Qualitative Research. *Educational Philosophy and Theory* 44, 4 (January 2012), 403–412. https://doi.org/10.1111/j.1469-5812.2010.00720.x

[14] Terrie E. Moffitt, Louise Arseneault, Daniel Belsky, Nigel Dickson, Robert J. Hancox, HonaLee Harrington, Renate Houts, Richie Poulton, Brent W. Roberts, Stephen Ross, Malcolm R. Sears, W. Murray Thomson, and Avshalom Caspi. 2011. A gradient of childhood self-control predicts health, wealth, and public safety. *Proc. Natl. Acad. Sci. U.S.A.* 108, 7 (February 2011), 2693–2698. https://doi.org/10.1073/pnas.1010076108

[15] Yusuke Moriguchi. 2014. The early development of executive function and its relation to social interaction: a brief review. *Front. Psychol.* 5, (April 2014). https://doi.org/10.3389/fpsyg.2014.00388

[16] J. Nielsen. 1993. Iterative user-interface design. *Computer* 26, 11 (November 1993), 32–41. https://doi.org/10.1109/2.241424

[17] Pantelis Pergantis, Victoria Bamicha, Charalampos Skianis, and Athanasios Drigas. 2025. AI Chatbots and Cognitive Control: Enhancing Executive Functions Through Chatbot Interactions: A Systematic Review. *Brain Sciences* 15, 1 (January 2025), 47. https://doi.org/10.3390/brainsci15010047

[18] Christian Roth and Hartmut Koenitz. 2016. Evaluating the User Experience of Interactive Digital Narrative. In *Proceedings of the 1st International Workshop on Multimedia Alternate Realities*, October 16, 2016. ACM, Amsterdam The Netherlands, 31–36. https://doi.org/10.1145/2983298.2983302

[19] Gabriela Sá, Cristina Sylla, Vítor Martins, Ana Caruso, and Douglas Menegazzi. 2019. Multiculturalism and Creativity in Storytelling - Visual Development of a Digital Manipulative for Young Children. In *Proceedings of the 2019 on Creativity and Cognition*, June 13, 2019. ACM, San Diego CA USA, 369–381. https://doi.org/10.1145/3325480.3326571

[20] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *CoDesign* 4, 1 (March 2008), 5–18. https://doi.org/10.1080/15710880701875068

[21] Cristina Sylla, Maitê Gil, Douglas Menegazzi, Rafael Fernandes, Cristiana Araújo, and Nuno Martins. 2021. Once Upon a Time: A Kit of Tools for Reading and Telling Stories. In *10th International Conference on Digital and Interactive Arts*, October 13, 2021. ACM, Aveiro, Portugal Portugal, 1–7. https://doi.org/10.1145/3483529.3483685

[22] Julie Thompson Burdine, Sally Thorne, and Gurjit Sandhu. 2021. Interpretive description: A flexible qualitative methodology for medical education research. *Medical Education* 55, 3 (March 2021), 336–343. https://doi.org/10.1111/medu.14380